

Saginala Hari Krishna

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SUMMARY

Computer Science student skilled in Full-Stack Development, Data Structures & Algorithms, and intelligent software systems. Experienced in building scalable web applications using React.js, Node.js, Express.js, Python, FastAPI, and MySQL. Developed projects involving conversational AI systems, RAG pipelines, real-time applications, and database-driven platforms. Passionate about backend engineering, REST API development, responsive UI design, and emerging AI technologies.

TECHNICAL SKILLS

Languages	Python, Java, C, C++, JavaScript
Frontend	React.js, HTML, CSS, Responsive UI Design
Backend	Node.js, Express.js, FastAPI, REST APIs, MySQL, PostgreSQL
AI / ML	LLM Integration, RAG, Whisper, Vector Databases, AI Agents
Core CS	Data Structures & Algorithms, Operating Systems, Computer Networks
Tools	Git, GitHub, Firebase, Supabase, Figma, n8n, Wireshark

PROJECTS

AI Clinical Assistant – (Link) *Python, FastAPI, Gemma 3-4B, Vector Database*
Developed an AI-driven virtual clinical front-desk assistant capable of handling **appointment scheduling, rescheduling, and patient query management** through context-aware conversational interactions. Implemented **memory management, token optimization, vector-based retrieval pipelines, and contextual conversation tracking** using the open-source **Gemma 3-4B** model to deliver accurate and professional healthcare assistance while maintaining respectful conversational behavior.

EduPDF Portal – (Demo) *JavaScript, Supabase, PostgreSQL, Razorpay, Firebase*
Built and deployed a centralized academic resource platform used by **100+ students** for accessing branch-wise and year-wise study materials. Integrated **Razorpay payment gateway**, role-based authentication, protected routes, and admin dashboards for secure premium content access and scalable PDF management workflows.

Hand Gesture Gaming System – (Link) *Python, OpenCV, Computer Vision*
Engineered a real-time **hand gesture detection** gaming platform for college Tech Fest using computer vision and motion tracking algorithms. Developed interactive **Fruit Ninja and steering-controlled racing games** with optimized gesture recognition, collision handling, and low-latency frame processing for smooth live gameplay experiences.

Smart Store Management System – (Link) *React.js, Node.js, Express.js, MySQL*
Developed a full-stack smart canteen and inventory platform that reduced **manual bookkeeping and transaction management** in college stores. Implemented **student wallet systems, inventory monitoring, purchase tracking, admin dashboards, and automated sales reporting** using scalable REST APIs and secure authentication workflows.

HostelHelper – (Demo) *React.js, Firebase, Supabase*
Developed a student collaboration platform enabling hostel students to connect with students available outside campus for assistance requests. Implemented **real-time availability tracking, secure authentication, request workflows, and rating-based trust systems** to streamline student coordination and improve communication efficiency.

EXPERIENCE

DRDO Internship – AI/ML Intern *Python, Whisper, Speech Processing*
Working on AI-based audio and video transcription systems using **Whisper models** for multilingual speech-to-text conversion. Implementing **speaker diarization, waveform visualization, offline inference pipelines, and transcription optimization workflows** for scalable speech processing applications.

EDUCATION

Jawaharlal Nehru Technological University, Anantapur	2024 – 2027
B.Tech in Computer Science and Engineering	CGPA: 8.65
Sri Venkateswara Government Polytechnic, Tirupati	2021 – 2024
Diploma in Computer Engineering	Percentage: 94.3%

ACHIEVEMENTS & CERTIFICATIONS

- TCS iON Career Edge – Young Professional Certification (Click)
- Selected as Google Gemini Campus Ambassador.
- Oracle Certified Foundations Associate (Click)
- McKinsey.org Forward Program (Click)